

openGUTS Report

Project:

GUTS_RF_Gammarus_45_Noprefitbackgroundhazard_29052026

Project file:

No project file saved or loaded

Project description (optional):

No project description available

Software version:

openGUTS - 1.2

Date of report creation:

01/06/2026 00:50:10

Calibration

Calibration input data

Data set 1

File: Data edited manually

Description (optional):

Gammarus

Control group: 'Control'

Survival data of input data set 1:

Time [d]	Control	T1	T2	T3	T4	T5	T6	T7
0	80	20	20	30	20	30	20	20
0.042	80	20	20	30	20	16	2	0
0.083	80	20	20	30	20	16	1	0
0.16	80	20	20	30	20	6	0	0
0.33	80	20	20	30	20	1	0	0
1	80	20	20	30	15	0	0	0
1.208	79	20	20	30	14	0	0	0
2	78	20	20	30	11	0	0	0
3	78	20	20	30	10	0	0	0
4	78	19	20	30	10	0	0	0

Concentration data of input data set 1:

Time [d]	Control	T1	T2	T3	T4	T5	T6	T7
0	0	1	3	10	30	100	300	1000
0.2	0	1	2.9	9.6	28.8	96.1	288.3	960.9
0.4	0	0.9	2.8	9.2	27.7	92.3	277	923.2
0.6	0	0.9	2.7	8.9	26.6	88.7	266.1	887.1
0.8	0	0.9	2.6	8.5	25.6	85.2	255.7	852.4
1	0	0.8	2.5	8.2	24.6	81.9	245.7	819
1.2	0	0.8	2.4	7.9	23.6	78.7	236.1	787
1.4	0	0.8	2.3	7.6	22.7	75.6	226.8	756.2
1.6	0	0.7	2.2	7.3	21.8	72.7	218	726.6

1.8	0	0.7	2.1	7	20.9	69.8	209.4	698.1
2	0	0.7	2	6.7	20.1	67.1	201.2	670.8
2.2	0	0.6	1.9	6.4	19.3	64.5	193.4	644.5
2.4	0	0.6	1.9	6.2	18.6	61.9	185.8	619.3
2.6	0	0.6	1.8	6	17.9	59.5	178.5	595.1
2.8	0	0.6	1.7	5.7	17.2	57.2	171.5	571.8
3	0	0.5	1.6	5.5	16.5	54.9	164.8	549.4
3.2	0	0.5	1.6	5.3	15.8	52.8	158.4	527.9
3.4	0	0.5	1.5	5.1	15.2	50.7	152.2	507.2
3.6	0	0.5	1.5	4.9	14.6	48.7	146.2	487.4
3.8	0	0.5	1.4	4.7	14	46.8	140.5	468.3
4	0	0.4	1.3	4.5	13.5	45	135	450

Calibration settings

Calibration parameter settings for GUTS-RED-SD:

Parameter	Fit	Min	Max	Scale
kd	Yes	0.001641	1096	Log
mw	Yes	0.0001887	990	Norm
hb	Yes	1E-6	0.07	Norm
bw	Yes	2.634E-5	334.5	Log
Fs	No	1	1	Norm

Calibration parameter settings for GUTS-RED-IT:

Parameter	Fit	Min	Max	Scale
kd	Yes	0.001641	79.68	Log
mw	Yes	0.0001887	141.5	Log
hb	Yes	1E-6	0.07	Norm
bw	No	Inf	Inf	Norm
Fs	Yes	1.05	20	Log

Calibration results

Fitted parameters for GUTS-RED-SD:

Best fit parameter values and their 95% CI

kd: 1096 (107.1 - 1096*)

mw: 21.08 (17.74 - 22.58)

hb: 0.006558 (0.001735 - 0.01524)

bw: 0.1282 (0.091 - 0.1691)

* edge of 95% parameter CI has run into a boundary

(this may also affect CIs of other parameters)

Goodness of fit for calibration data (GUTS-RED-SD):

Model efficiency (NSE, r-square): 0.9733

Normalised root-means-square error (NRMSE): 7.408 %

Minus log-likelihood (MLL): 109.17

AIC: 226.34

Survival probability prediction error (SPPE) for each treatment:

Data set	Treatment	Value
1	Control	0.08916 %
1	T1	-2.41 %
1	T2	2.589 %
1	T3	2.589 %
1	T4	12.24 %
1	T5	-2.29E-9 %
1	T6	-5.23E-40 %
1	T7	-2.99E-147 %

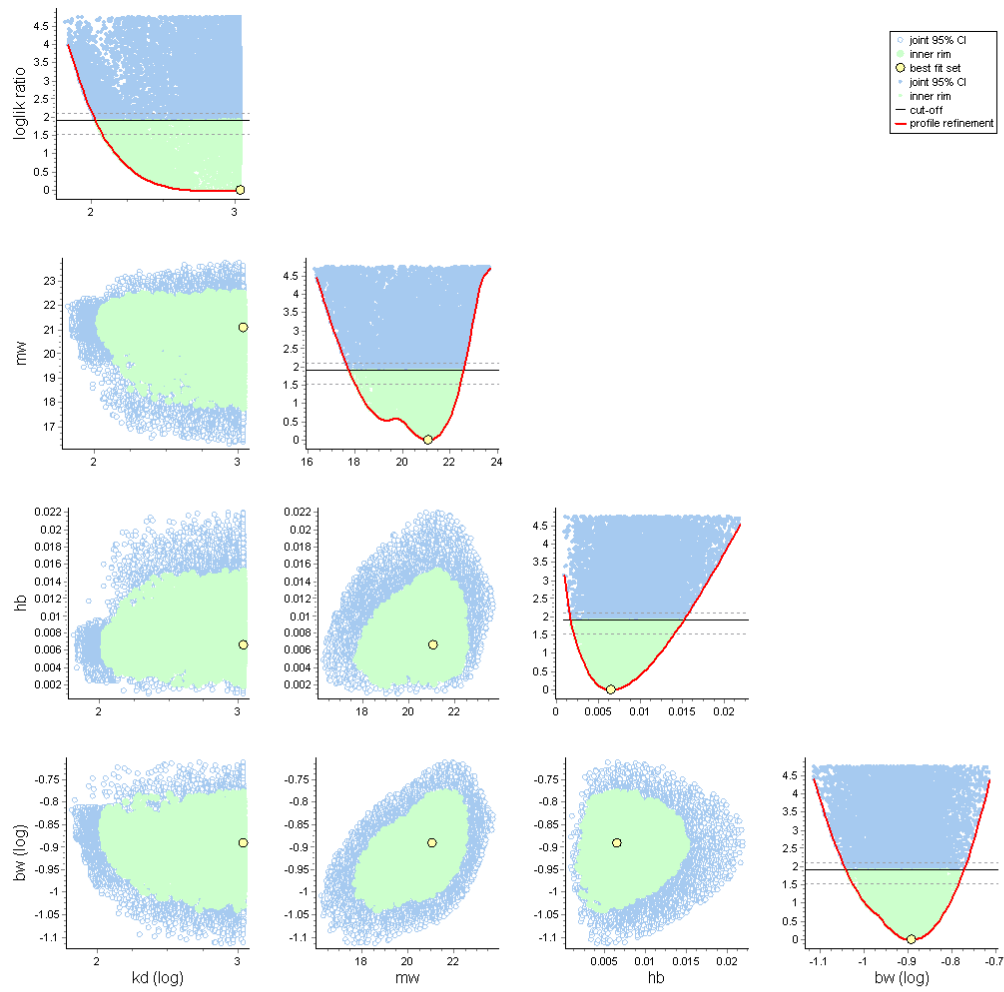
GUTS-RED-SD results table for LC_{x,t} [[C]], with 95% CI:

Time [d]	LC50	LC20	LC10
1	26.5 (23.81 - 28.47)	22.83 (19.75 - 24.29)	21.91 (18.66 - 23.37)
2	23.79 (20.87 - 25.28)	21.95 (18.71 - 23.41)	21.49 (18.15 - 22.96)

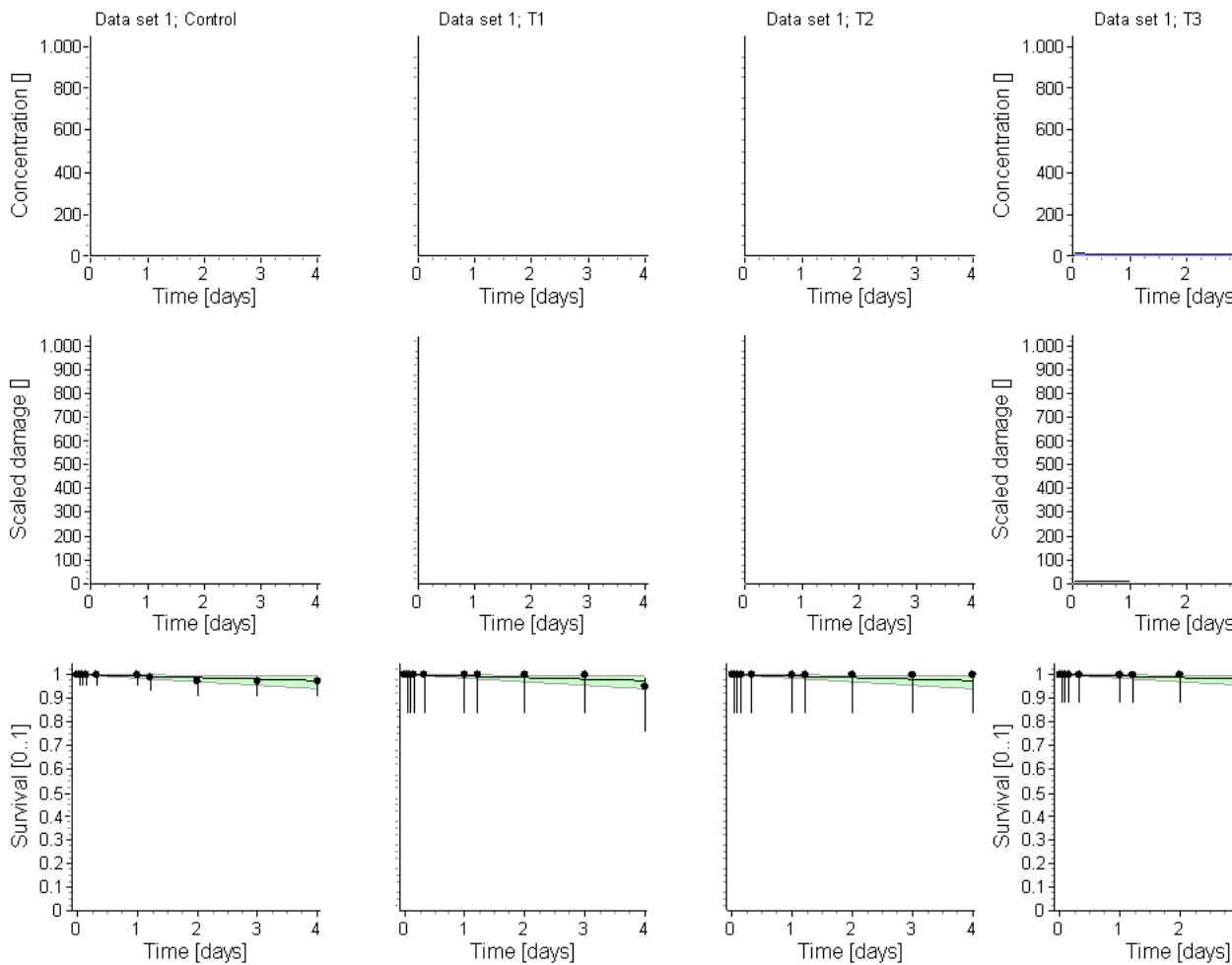
3	22.88 (19.82 - 24.35)	21.66 (18.36 - 23.12)	21.35 (17.99 - 22.83)
4	22.43 (19.29 - 23.9)	21.52 (18.18 - 22.98)	21.29 (17.9 - 22.77)
7	21.85 (18.59 - 23.31)	21.33 (17.96 - 22.81)	21.2 (17.8 - 22.69)
10	21.62 (18.31 - 23.08)	21.25 (17.87 - 22.74)	21.16 (17.76 - 22.66)
14	21.47 (18.12 - 22.94)	21.2 (17.81 - 22.69)	21.14 (17.73 - 22.63)
21	21.34 (17.97 - 22.82)	21.16 (17.76 - 22.66)	21.12 (17.7 - 22.62)
27	21.28 (17.9 - 22.76)	21.15 (17.73 - 22.64)	21.11 (17.69 - 22.61)
28	21.27 (17.89 - 22.76)	21.14 (17.73 - 22.64)	21.11 (17.69 - 22.61)
42	21.21 (17.81 - 22.7)	21.12 (17.71 - 22.62)	21.1 (17.68 - 22.6)
50	21.19 (17.79 - 22.68)	21.12 (17.7 - 22.61)	21.1 (17.68 - 22.59)
100	21.13 (17.72 - 22.63)	21.1 (17.68 - 22.6)	21.09 (17.67 - 22.59)
182	21.11 (17.69 - 22.61)	21.09 (17.67 - 22.59)	21.09 (17.66 - 22.58)

Plots for GUTS-RED-SD calibration:

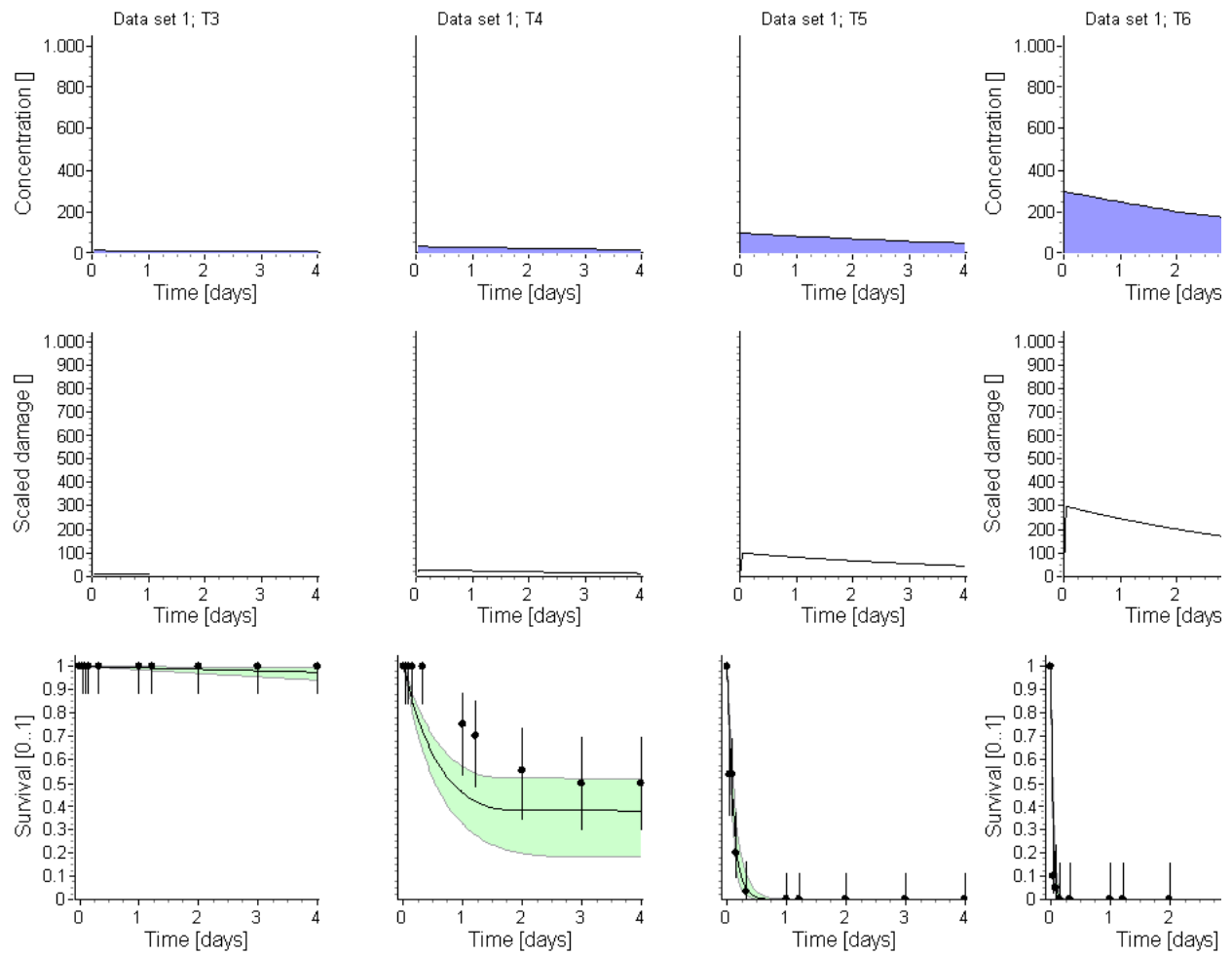
Parameter space plot for the calibration of GUTS-RED-SD:



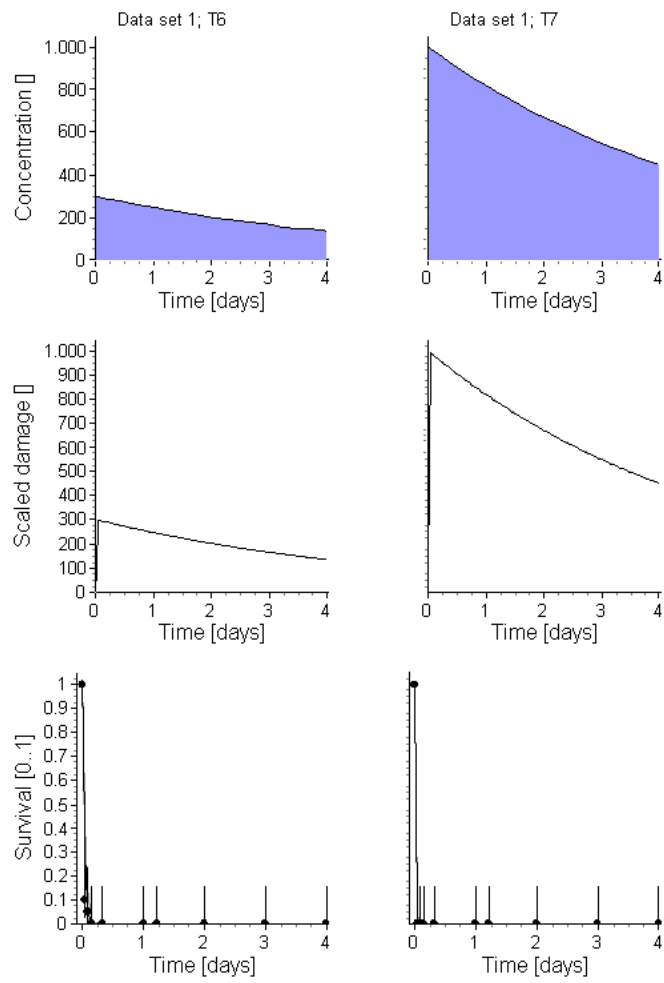
Exposure, damage and survival plots for the calibration of GUTS-RED-SD:



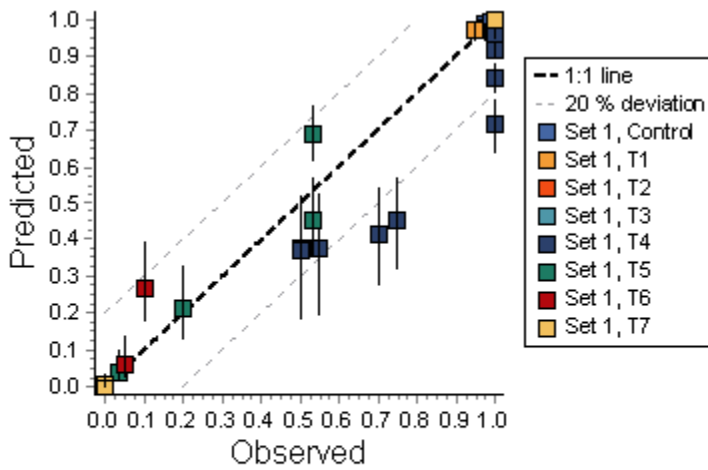
... continued plot:



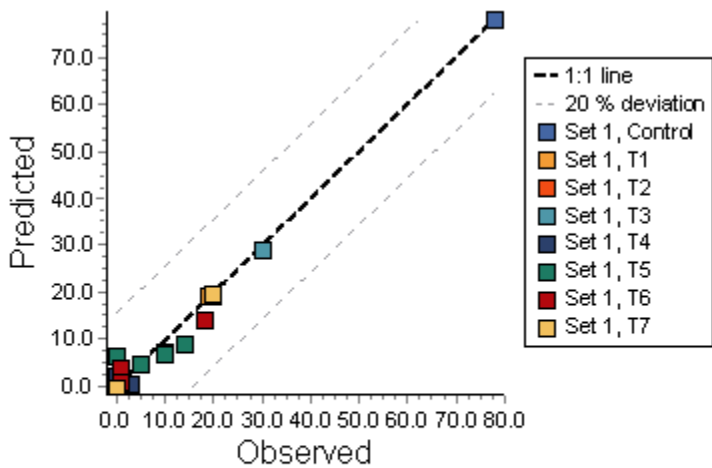
... continued plot:



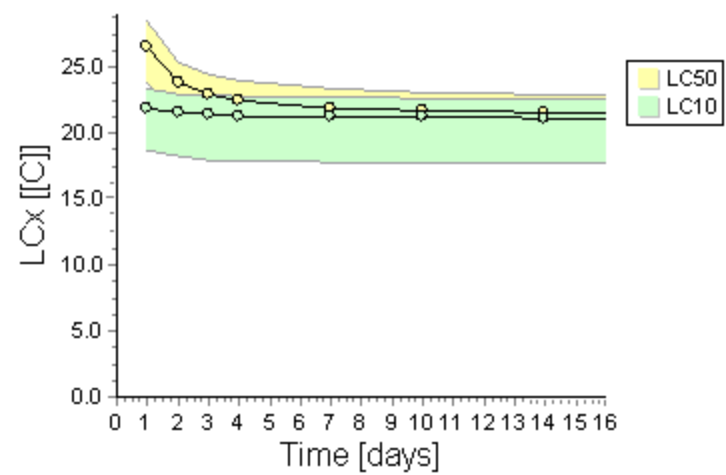
Observed vs. Predicted survival plot for the calibration of GUTS-RED-SD:



Observed vs. Predicted deaths plot for the calibration of GUTS-RED-SD:



LCx versus time with confidence intervals (plotted for 16 days, GUTS-RED-SD):



Fitted parameters for GUTS-RED-IT:

Best fit parameter values and their 95% CI

kd: 4.672 (3.181 - 6.673)

mw: 30.8 (23.22 - 40.36)

hb: 0.01293 (0.005915 - 0.02408)

Fs: 3.907 (2.875 - 5.983)

* edge of 95% parameter CI has run into a boundary

(this may also affect CIs of other parameters)

Goodness of fit for calibration data (GUTS-RED-IT):

Model efficiency (NSE, r-square): 0.9799

Normalised root-means-square error (NRMSE): 7.595 %

Minus log-likelihood (MLL): 127.71

AIC: 263.41

Survival probability prediction error (SPPE) for each treatment:

Data set	Treatment	Value
1	Control	2.542 %
1	T1	0.04856 %
1	T2	5.17 %
1	T3	8.105 %
1	T4	-7.97 %
1	T5	-5.51 %
1	T6	-0.3 %
1	T7	-0.01 %

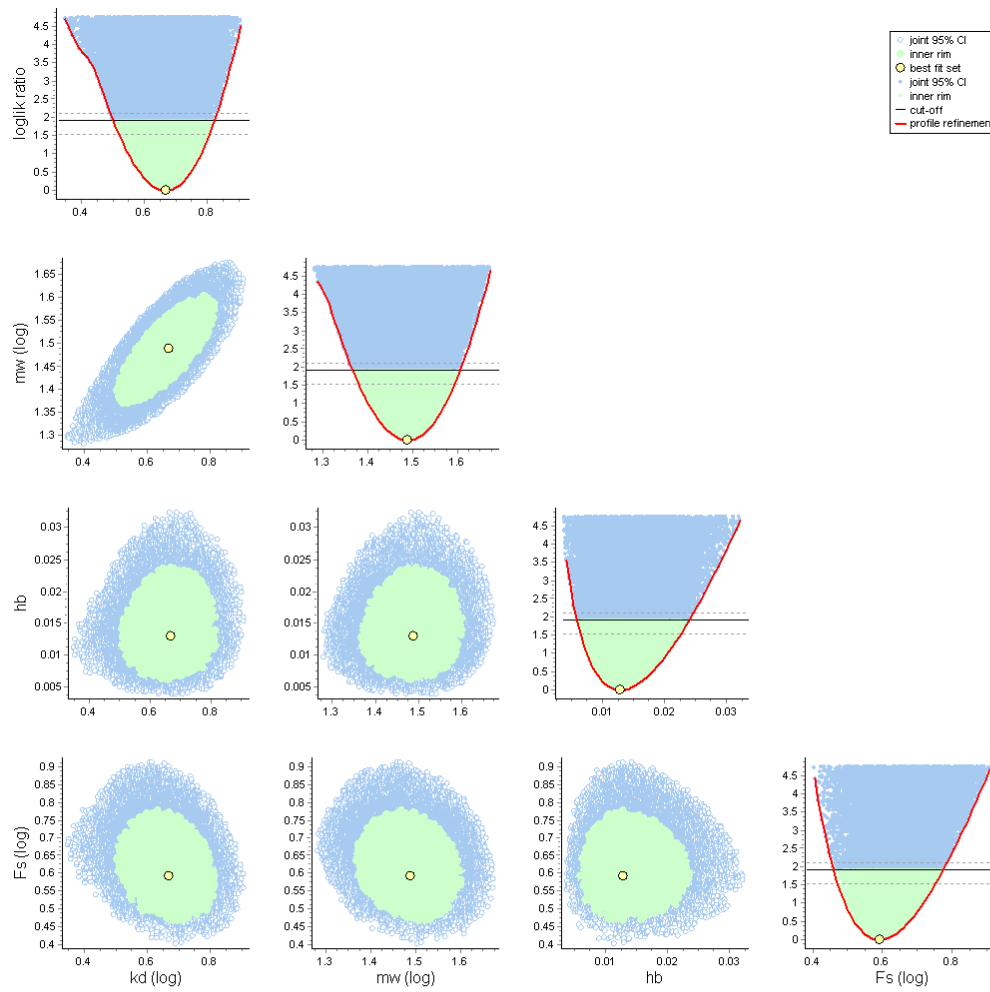
GUTS-RED-IT results table for LC_{x,t} [[C]], with 95% CI:

Time [d]	LC50	LC20	LC10
1	31.09 (23.72 - 40.76)	18.56 (13.14 - 24.83)	13.73 (9.071 - 19.27)
2	30.8 (22.96 - 40.68)	18.39 (12.81 - 24.74)	13.6 (8.872 - 19.2)
3	30.8 (22.93 - 40.68)	18.39 (12.8 - 24.74)	13.6 (8.867 - 19.2)
4	30.8 (22.93 - 40.68)	18.39 (12.8 - 24.74)	13.6 (8.867 - 19.2)

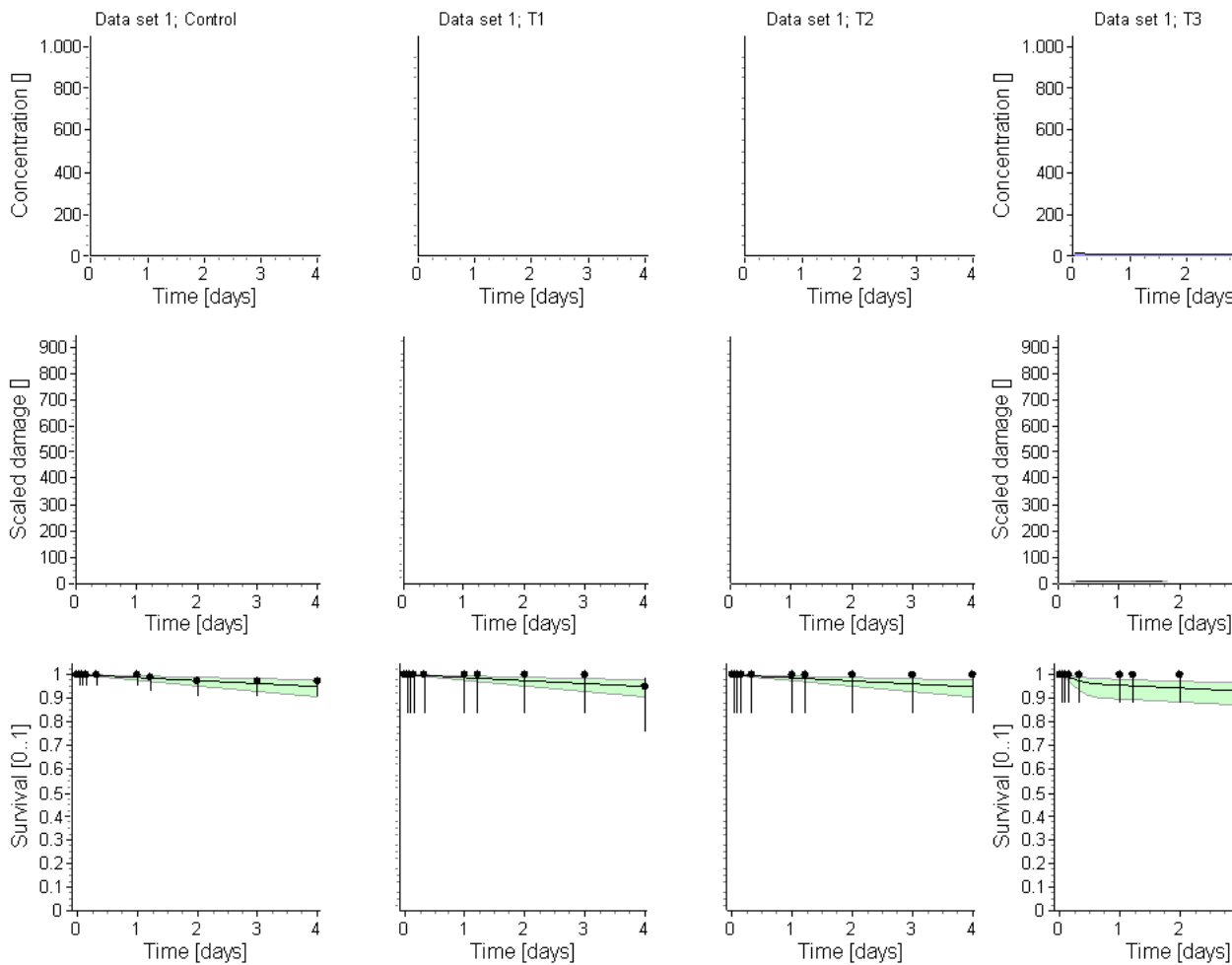
7	30.8 (22.93 - 40.68)	18.39 (12.8 - 24.74)	13.6 (8.867 - 19.2)
10	30.8 (22.93 - 40.68)	18.39 (12.8 - 24.74)	13.6 (8.867 - 19.2)
14	30.8 (22.93 - 40.68)	18.39 (12.8 - 24.74)	13.6 (8.867 - 19.2)
21	30.8 (22.93 - 40.68)	18.39 (12.8 - 24.74)	13.6 (8.867 - 19.2)
27	30.8 (22.93 - 40.68)	18.39 (12.8 - 24.74)	13.6 (8.867 - 19.2)
28	30.8 (22.93 - 40.68)	18.39 (12.8 - 24.74)	13.6 (8.867 - 19.2)
42	30.8 (22.93 - 40.68)	18.39 (12.8 - 24.74)	13.6 (8.867 - 19.2)
50	30.8 (22.93 - 40.68)	18.39 (12.8 - 24.74)	13.6 (8.867 - 19.2)
100	30.8 (22.93 - 40.68)	18.39 (12.8 - 24.74)	13.6 (8.867 - 19.2)
182	30.8 (22.93 - 40.68)	18.39 (12.8 - 24.74)	13.6 (8.867 - 19.2)

Plots for GUTS-RED-IT calibration:

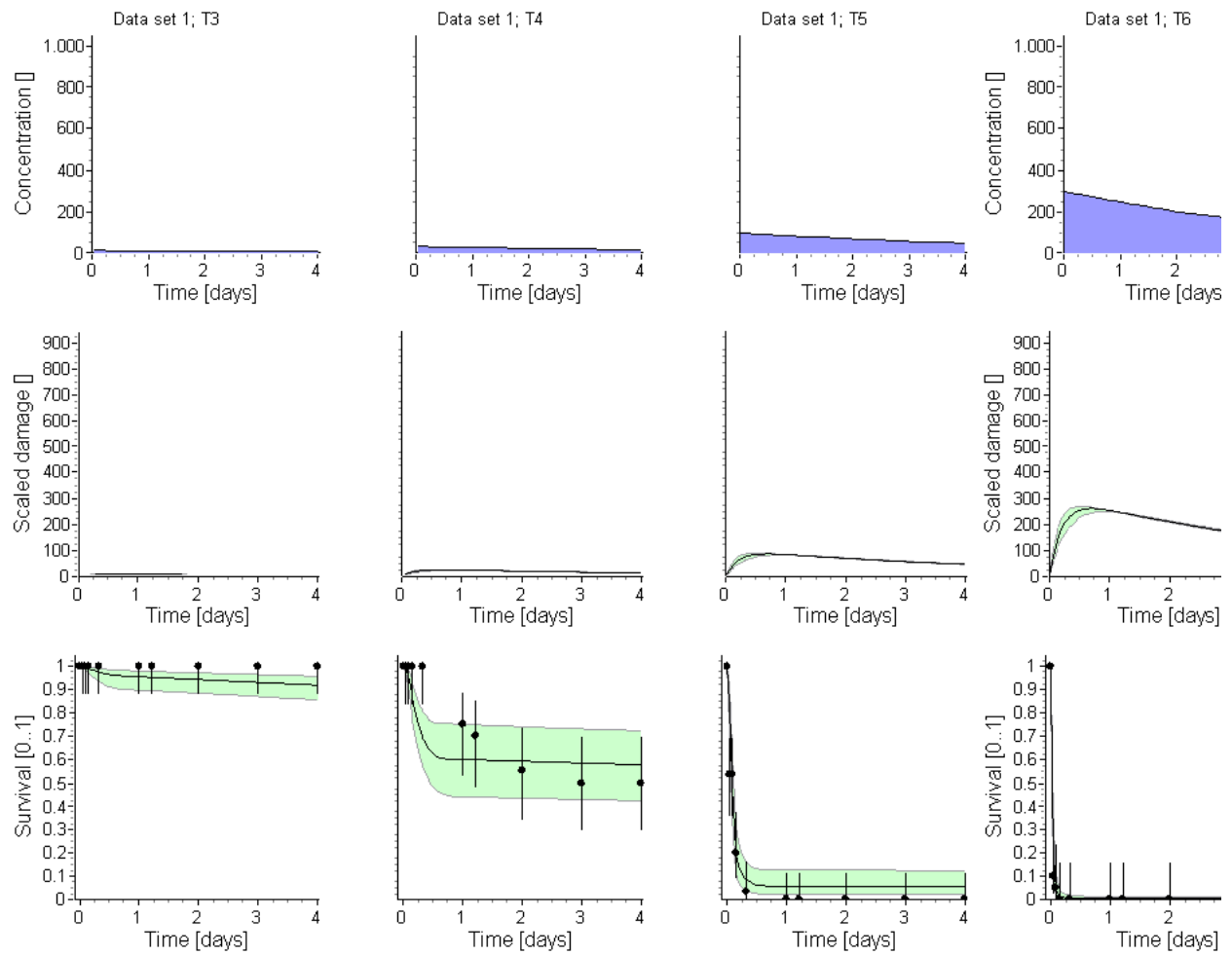
Parameter space plot for the calibration of GUTS-RED-IT:



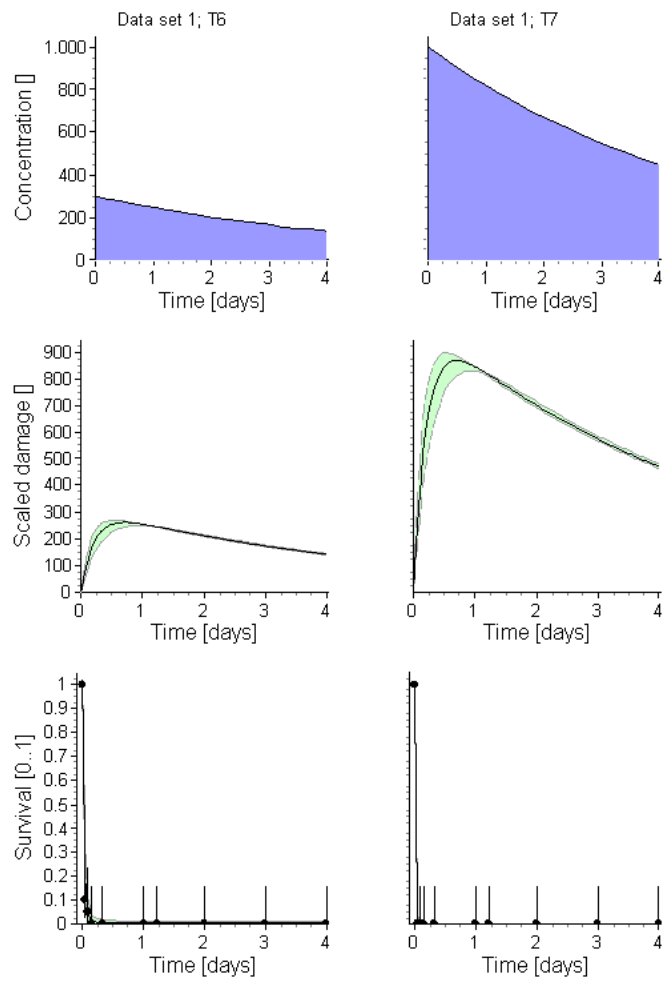
Exposure, damage and survival plots for the calibration of GUTS-RED-IT:



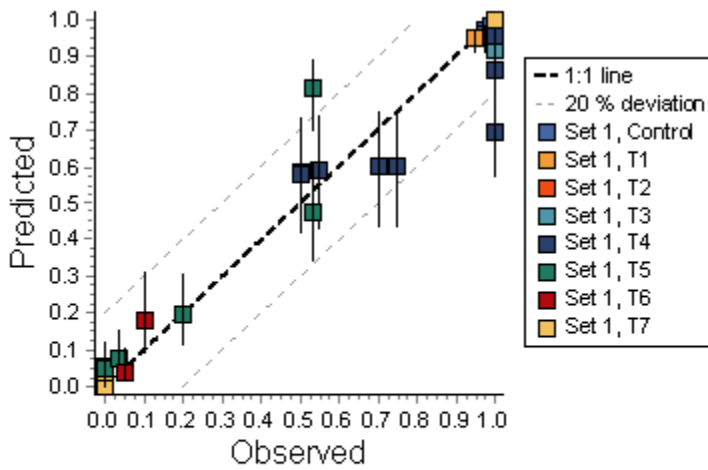
... continued plot:



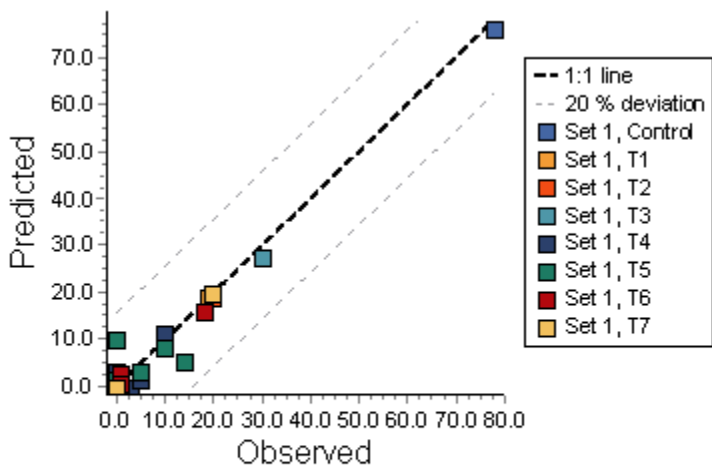
... continued plot:



Observed vs. Predicted survival plot for the calibration of GUTS-RED-IT:



Observed vs. Predicted deaths plot for the calibration of GUTS-RED-IT:



LCx versus time with confidence intervals (plotted for 16 days, GUTS-RED-IT):

Validation

No validation performed!

Predictions

No predictions performed!